Models of Expectations Formation

Information, Rationality, Heterogeneity, and Equilibrium

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Organizing Principle: Separate FI from RE

- **FI (Full Information):** what does the agent observe?
- **RE (Rational Expectations):** how does the agent map information into beliefs?
  - Rationality *conditional on* the information held
- Two independent assumptions; their violations need not coincide

$$\underbrace{F_t^i x_{t+h}}_{\text{subjective forecast}} = \underbrace{\mathcal{B}_i}_{\text{belief operator}} \Big(\underbrace{\mathcal{I}_{it}}_{\text{information set}} \Big)$$

$$\text{FIRE:} \qquad \mathcal{I}_{it} = \mathcal{I}_t^{\text{full}}, \qquad \mathcal{B}_i(\mathcal{I}_{it}) = \mathbb{E}[x_{t+h} \mid \mathcal{I}_{it}]$$

- **Key papers:** Muth 61; Lucas 72; Coibion–Gorodnichenko 15

A Two-by-Two Taxonomy

	RE	**Non-RE**
Full Information	**I. FIRE benchmark** Canonical RE / NK	**III. Behavioral beliefs** Diagnostic, extrapolative, memory
Imperfect Information	**II. Information frictions** Sticky / noisy information; rational inattention	**IV. Joint distortions** Diagnostic Kalman filter; biased updating

- Quadrants are combinations of assumptions, not mutually exclusive literatures
- Common test: forecast errors on forecast revisions

$$\underbrace{x_{t+h} - F_t x_{t+h}}_{\mathrm{FE}_t} = \alpha + \beta \underbrace{\left(F_t x_{t+h} - F_{t-1} x_{t+h}\right)}_{\mathrm{FR}_t} + e_t, \qquad \begin{cases} \beta = 0 & \text{FIRE} \\ \beta > 0 & \text{underreaction} \\ \beta < 0 & \text{overreaction} \end{cases}$$

- Consensus $\bar{F}$ tests FI; individual F^i tests RE

Data and Estimation Toolkit

Survey	Variables	Level / frequency
SPF (Philadelphia Fed)	GDP, inflation, rates; density forecasts	Forecaster, Q, since 1968
Blue Chip	GDP, inflation, Treasury, AAA/BAA	Forecaster, M (proprietary)
ECB SPF	Euro-area inflation, GDP; densities	Forecaster, Q, since 1999
Michigan	Inflation, sentiment	Household, M (not panel)
NY Fed SCE	Inflation, income, house prices	Household panel, M, since 2013
Duke CFO Survey	Earnings, capex, wages, S&P returns	Firm, Q, since 1998
IBES	EPS, sales, LTG	Analyst, since 1980s
Gallup / AAII	Stock market expectations	Aggregate, M / W

- **Estimation pitfalls**
  - Overlapping horizons $\Rightarrow$ serially correlated errors; use Newey–West / Driscoll–Kraay
  - Small-T bias with persistent regressors (Stambaugh, Nickell): $\mathbb{E}[\widehat{\rho} - \rho] \approx -(1+3\rho)/T$
  - i.i.d. process $\Rightarrow \mathrm{FR} = 0$ under RE, β not identified

I. Full Information Rational Expectations

- **Assumptions**
  - Economy-wide state z_t is observed; true transition law is used
- **Mechanism:** forecast errors are orthogonal to time-t information

$$z_{t+1} = Az_t + u_{t+1}, \qquad \mathbb{E}_t z_{t+h} = A^h z_t, \qquad \mathbb{E}\left[z_{t+h} - \mathbb{E}_t z_{t+h} \mid \mathcal{I}_t\right] = 0$$

- **Empirics**
  - Data: CFO Survey, IBES, Gallup, Vanguard accounts
  - Estimation: regress decisions on survey expectations, controlling for Q / fundamentals
  - Values: expectations predict investment and fund flows beyond standard predictors
- **Key papers:** Greenwood–Shleifer 14; Gennaioli–Ma–Shleifer 16; Giglio–Maggiori–Stroebel–Utkus 21

II-A. Sticky Information

- **Updating**
  - Update with probability $1-\lambda$; RE conditional on updating
- **Predictions**
  - Individual: errors unpredictable by own revisions
  - Consensus: revisions insufficient, $\beta > 0$

$$\bar{F}_t x_{t+h} = (1-\lambda)\sum_{j=0}^{\infty} \lambda^j \, \mathbb{E}_{t-j} x_{t+h} \quad \Longrightarrow \quad \beta = \frac{\lambda}{1-\lambda}, \qquad \lambda = \frac{\beta}{1+\beta}$$

- **Empirics**
  - Data: SPF, Michigan
  - Estimation: consensus OLS of FE on FR; map $\widehat{\beta}$ to λ
  - Values: $\lambda = 0.75$ (quarterly; one update per year) in Mankiw–Reis 02; inflation $\widehat{\beta} \approx 1.2 \Rightarrow \lambda \approx 0.55$
- **Key papers:** Mankiw–Reis 02; Carroll 03; Coibion–Gorodnichenko 15

II-B. Noisy Information

- **Signals:** private and noisy; optimal Kalman filtering
- **Predictions**
  - Individual: errors unpredictable
  - Consensus: underreacts to news; disagreement despite RE

$$s_{it} = z_t + \varepsilon_{it}, \qquad F^i_t z_t = G\, s_{it} + (1-G)\, F^i_{t-1} z_t, \qquad G = \frac{\Sigma}{\Sigma + \sigma^2_\varepsilon}$$

$$\bar{F}_t z_{t+h} = G\, \mathbb{E}_t z_{t+h} + (1-G)\, \bar{F}_{t-1} z_{t+h} \quad \Longrightarrow \quad \beta = \frac{1-G}{G}$$

- **Empirics**
  - Data: SPF, Blue Chip; identified shocks (oil, TFP, monetary)
  - Estimation: (i) consensus FE on FR; (ii) IRF of consensus errors to shocks
  - Values: inflation $\widehat{\beta} \approx 1.2 \Rightarrow G \approx 0.45$; errors converge to zero only gradually after shocks
- **Key papers:** Woodford 03; Coibion–Gorodnichenko 12, 15

II-C. Rational Inattention

- **Information is chosen**
  - Agents choose how much information to process; filtering is rational
- **Mechanism:** attention shifts with volatility, incentives, and policy regime

$$\min_{q(s|z),\, a(s)} \mathbb{E}\Big[L(a,z)\Big] + \mu\, I(z;s), \qquad \text{Gaussian: } G = 1 - 2^{-2\kappa}$$

- **Empirics**
  - Data: sectoral prices (CPI micro); firm / household attention surveys
  - Estimation: match price IRFs to aggregate vs. idiosyncratic shocks
  - Values: firms attend mostly to idiosyncratic shocks $\Rightarrow$ fast sectoral, slow aggregate price responses
- **Key papers:** Sims 03; Maćkowiak–Wiederholt 09; Maćkowiak–Matějka–Wiederholt 23

III-A. Diagnostic Expectations

- **Representativeness:** overweight states that became more likely given news
- **Properties:** kernel of truth; RE nested at $\theta = 0$, $\rho = 0$, or no news

$$h_t^\theta(\omega_{t+1}) \;\propto\; h(\omega_{t+1} \mid \omega_t) \left[\frac{h(\omega_{t+1} \mid \omega_t)}{h(\omega_{t+1} \mid \rho\, \omega_{t-1})} \right]^\theta$$

$$\mathbb{E}_t^\theta \omega_{t+1} = \mathbb{E}_t \omega_{t+1} + \rho\theta\epsilon_t, \qquad \mathbb{E}_t\left[\omega_{t+1} - \mathbb{E}_t^\theta \omega_{t+1}\right] = -\rho\theta\epsilon_t$$

- **Empirics**
  - Data: Blue Chip credit spreads; IBES long-term growth (LTG); SPF / Blue Chip individual
  - Estimation: individual FE on FR; SMM matching β across series
  - Values: $\theta \approx 0.5$–1; LTG $\beta \approx -0.3$
- **Key papers:** Gennaioli–Shleifer 10; Bordalo–Gennaioli–Shleifer 18; Bordalo–Gennaioli–La Porta–Shleifer 19

III-B. Extrapolation and Learning

- **Mechanical extrapolation:** no kernel of truth; ϕ must vary across settings
- **Adaptive learning:** agents estimate a perceived law of motion
  - Learning $\neq$ irrationality; classification depends on the benchmark

$$F_t x_{t+1} = x_t + \phi(x_t - x_{t-1}), \qquad \widehat{\rho}_t = \widehat{\rho}_{t-1} + g_t\, R_t^{-1} x_{t-1}\left(x_t - \widehat{\rho}_{t-1} x_{t-1}\right)$$

$$\text{constant gain: } g_t = g, \qquad \text{experience-based: } g_{t,\text{age}} = \vartheta/\text{age}$$

- **Empirics**
  - Data: Gallup, AAII, Michigan (cohort $\times$ time)
  - Estimation: expectations on past returns / own-lifetime inflation; NLS for gain
  - Values: expectations load positively on past returns, predict low future returns; $\widehat{\vartheta} \approx 3$
- **Key papers:** Greenwood–Shleifer 14; Evans–Honkapohja 01; Malmendier–Nagel 16

III-C. Memory and Long-Run Beliefs

- **Imperfect memory** overweights recent observations
- **Biased long-run mean:** bias rises when $\rho \downarrow$ and $h \uparrow$
  - Fits stronger overreaction for transitory processes and long horizons

$$\widehat{\mu}_t = \mathbb{E}[\mu \mid m_t] + \gamma_t\big(x_t - (1-\rho)\mathbb{E}[\mu \mid m_t] - \rho x_{t-1}\big), \qquad F_t x_{t+h} = (1-\rho^h)\,\widehat{\mu}_t + \rho^h x_t$$

- **Empirics**
  - Data: lab forecasts of simulated AR(1), randomized $\rho \in \{0, \ldots, 1\}$
  - Estimation: implied persistence $F_t x_{t+1} = \alpha + \widehat{\rho}\, x_t$; individual FE on FR
  - Values: implied $\widehat{\rho} = 0.45, 0.62, 0.77, 0.85, 0.91, 0.99$ for $\rho = 0, 0.2, \ldots, 1$; overreaction strongest at low ρ
- **Key papers:** da Silveira–Sung–Woodford 20; Afrouzi–Kwon–Landier–Ma–Thesmar 23

IV. Noisy Information with Biased Updating

- **Two layers:** noisy private signals and distorted posterior
- **Aggregation reverses the sign:** individual $\beta < 0$, consensus β may be > 0

$$x^{i,\theta}_{t+h|t} = x^{i}_{t+h|t-1} + (1+\theta)\frac{\Sigma}{\Sigma+\sigma^2_\epsilon}\rho^h\left(s^i_t - x^i_{t|t-1}\right)$$

$$\text{sign}(\beta) = \text{sign}\left(\sigma^2_\epsilon - \theta\Sigma\right), \qquad \beta_p = -\frac{\theta(1+\theta)}{(1+\theta)^2+\theta^2\rho^2} < 0$$

- **Empirics**
  - Data: SPF and Blue Chip, 22 series, individual and consensus
  - Estimation: jointly match individual β_p and consensus β for $(\theta, \sigma_\epsilon)$
  - Values: individual $\beta_p < 0$ for most series; consensus $\beta > 0$; exception: short rates ($\beta_p > 0$)
- **Key papers:** Bordalo–Gennaioli–Ma–Shleifer 20; Angeletos–Huo–Sastry 21

A Separate Axis: Higher-Order Beliefs

- Agents care about what others expect; complementarity amplifies frictions
- An **equilibrium layer**, not a fifth FIRE quadrant

$$p_{it} = \mathbb{E}_{it}\left[(1-\alpha)m_t + \alpha p_t\right] \quad \Longrightarrow \quad p_t = (1-\alpha)\sum_{k=0}^{\infty} \alpha^k \, \bar{\mathbb{E}}_t^{(k+1)} m_t$$

- **Empirics**
  - Data: rarely observed directly; inferred from IRFs or from consensus vs. individual gaps
  - Estimation: Bayesian estimation of dispersed-information DSGE; IRF matching
  - Values: complementarity $\alpha = 0.85$ ($\xi = 0.15$) in Woodford 03 $\Rightarrow$ higher orders dominate
- **Key papers:** Morris–Shin 02; Woodford 03; Angeletos–Lian 18

A Separate Axis: Heterogeneous Beliefs

- **Sources:** information sets, priors, models, attention
- **Disagreement vs. uncertainty**
  - Disagreement: dispersion of point forecasts across agents
  - Uncertainty: dispersion within one agent's distribution

$$D_t = \mathrm{Var}_i\left(\mathbb{E}_{it} x_{t+h}\right), \qquad U_{it} = \mathrm{Var}_{it}(x_{t+h}), \qquad \mathrm{Var}(X) = \mathbb{E}_i[U_i] + \mathrm{Var}_i(\mu_i)$$

- **Empirics**
  - Data: SPF / ECB SPF density bins; NY Fed SCE; Michigan; Vanguard
  - Estimation: IQR of point forecasts; fit generalized beta to histograms for U_{it}
  - Values: disagreement and uncertainty only weakly correlated; heterogeneity mostly persistent individual effects
- **Key papers:** Mankiw–Reis–Wolfers 03; Engelberg–Manski–Williams 09; Giglio et al. 21

From Belief Models to Macroeconomic Outcomes

1. **Micro:** choose information and form a forecast
2. **Decision:** price setting, consumption, investment, portfolio choice
3. **Aggregation:** strategic interaction, heterogeneous exposures
4. **General equilibrium:** inflation, output, yields, welfare
5. **Policy:** stabilization under endogenous expectations

$$\mathcal{I}_{it} \;\to\; F_t^i \;\to\; a_{it} \;\to\; (\pi_t, y_t, r_t) \;\to\; \mathcal{L}, \qquad \pi_t = \beta\, \bar{F}_t \pi_{t+1} + \kappa\, y_t + u_t$$

- **Empirics**
  - Data: SPF / Michigan inflation expectations in place of $\mathbb{E}_t \pi_{t+1}$
  - Estimation: OLS / GMM of NKPC with survey expectations
  - Values: survey-based NKPC fits better than RE version; household expectations explain missing disinflation
- **Key papers:** Adam–Padula 11; Coibion–Gorodnichenko 15b; Coibion–Gorodnichenko–Kamdar 18

Modular Model Note: One Model, One Entry

- **Assumptions**
  - Information set, objective DGP, subjective belief operator
- **Mechanism**
  - Which friction changes which decision margin?
- **Equilibrium**
  - How do heterogeneous decisions aggregate?
- **Empirics**
  - Data, estimation, key parameter values
- **Policy**
  - Welfare implications and counterfactual benchmark

$$\text{Assumptions} \rightarrow \text{Beliefs} \rightarrow \text{Actions} \rightarrow \text{Equilibrium} \rightarrow \text{Policy}$$

What to Add Next?

- **Alternative priors and ambiguity**
  - Robust control, ambiguity-sensitive decisions
- **Learning dynamics**
  - Bayesian vs. adaptive learning, model uncertainty
- **Endogenous information**
  - Costly acquisition, communication, asymmetric attention
- **Social and higher-order beliefs**
  - Networks, coordination, dispersed information
- **Subjective distributions**
  - Uncertainty, skewness, tail beliefs
- Independent modelling dimensions layered on the FIRE grid, not new quadrants